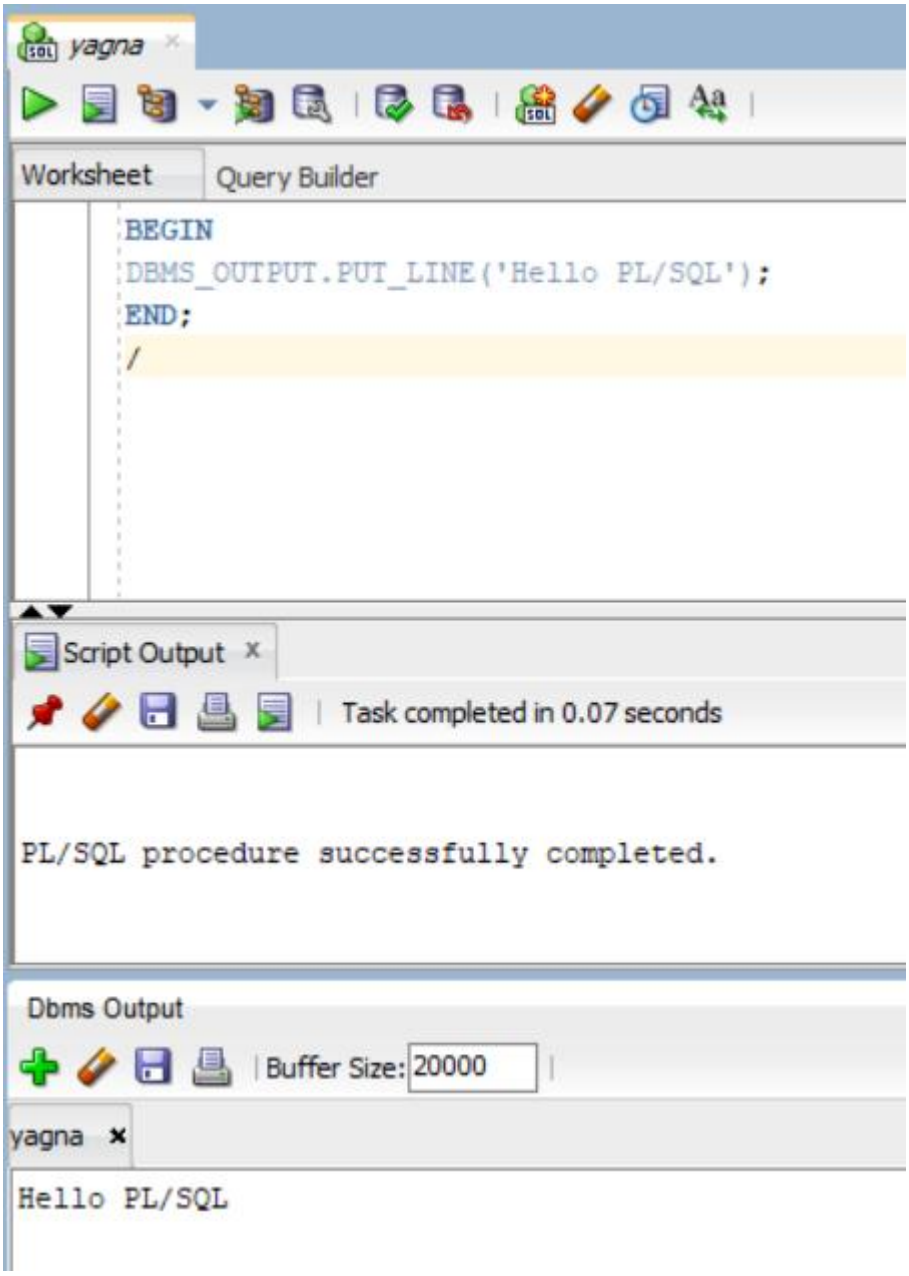
Module 3 - pl/sql programming

1. Hello program

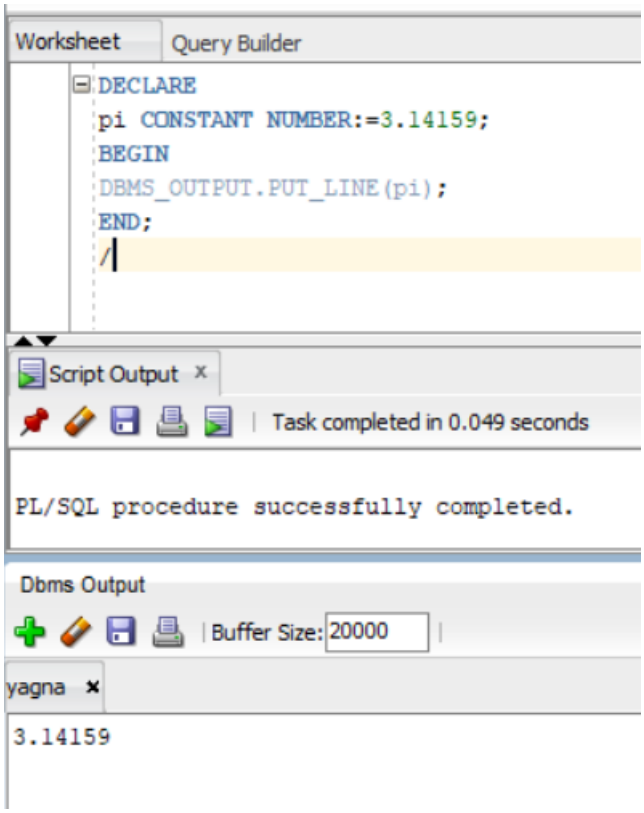


2. Variables & data types

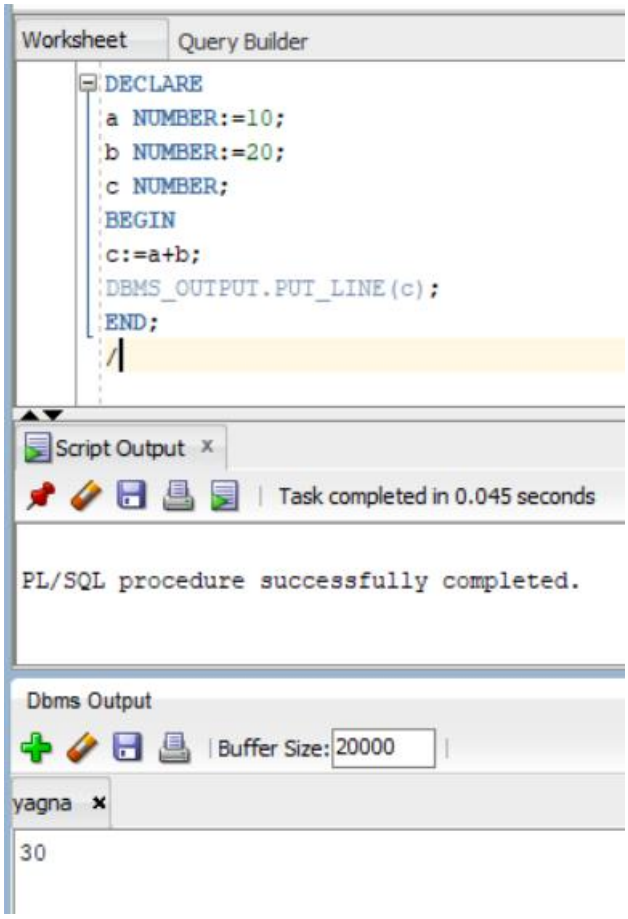
```
DECLARE
name VARCHAR2(20):='Yagna';
age NUMBER:=21;
BEGIN
DBMS_OUTPUT.PUT_LINE(name);
DBMS_OUTPUT.PUT_LINE(age);
END;
/
```

```
Yagna
21
```

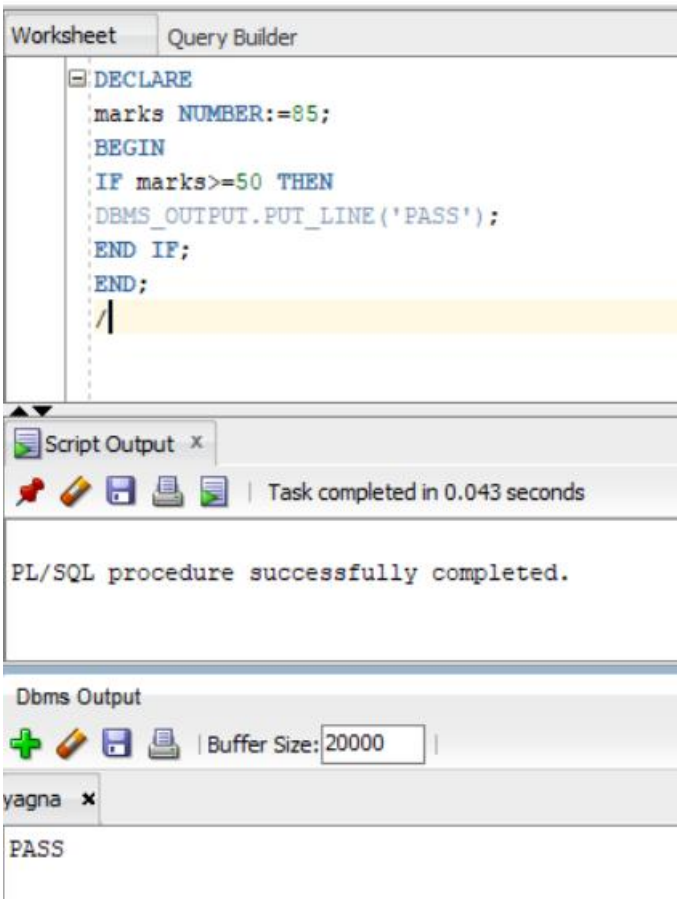
3. Constants



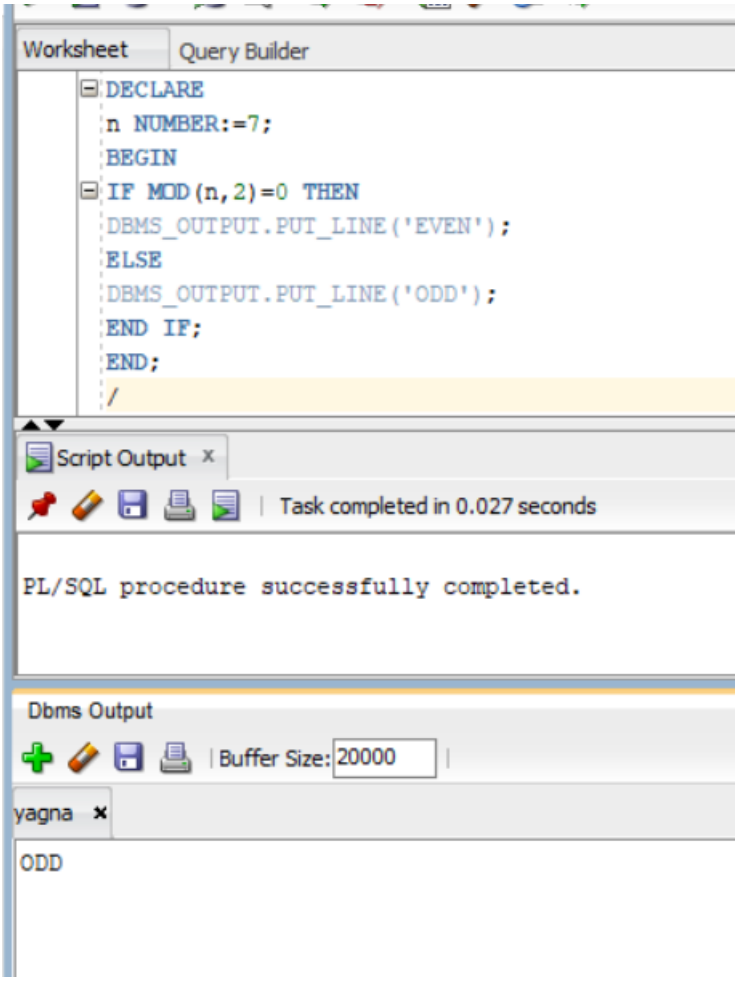
4. Accepting variable values



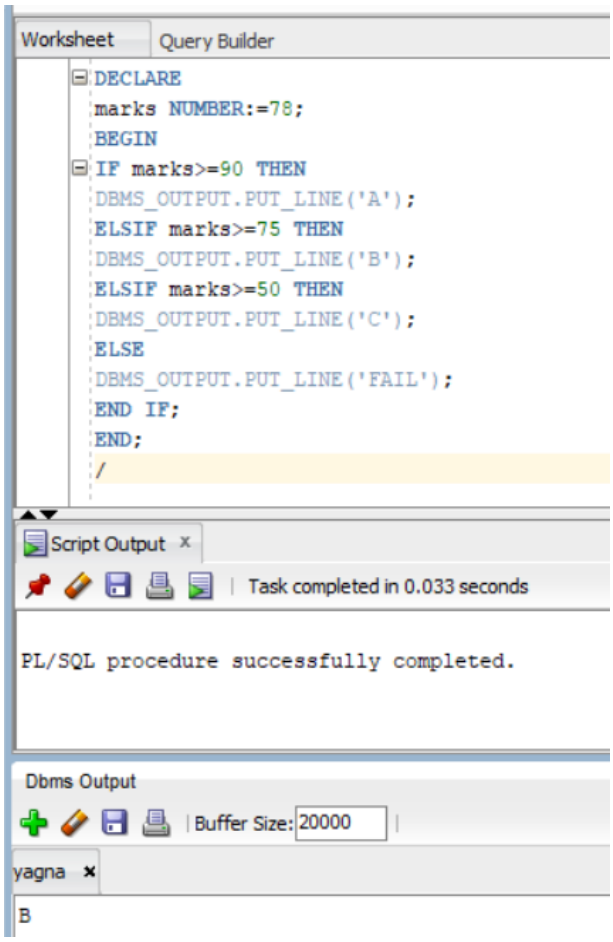
5. If statement



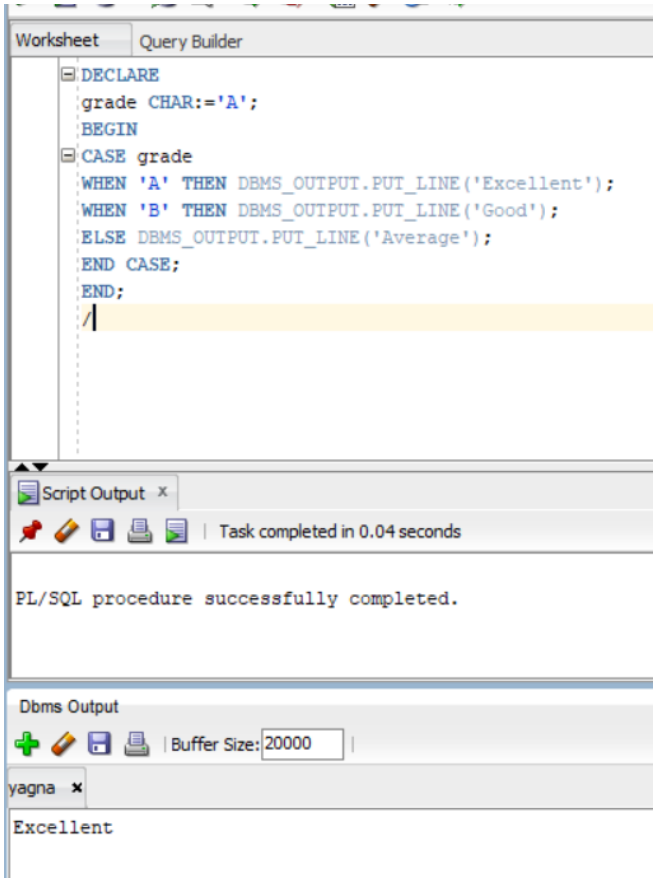
6. If else



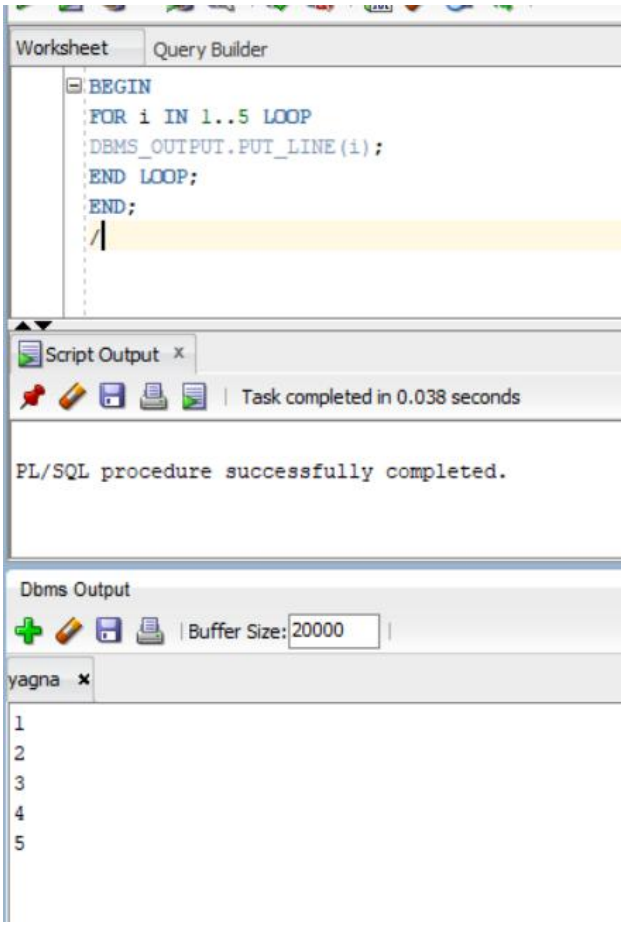
7. If elsif else



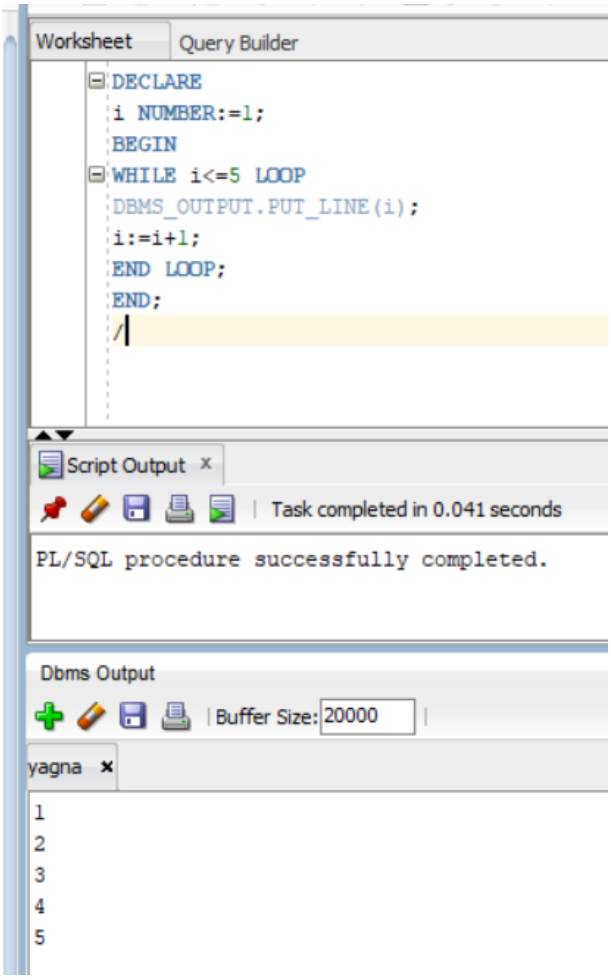
8. Case statement



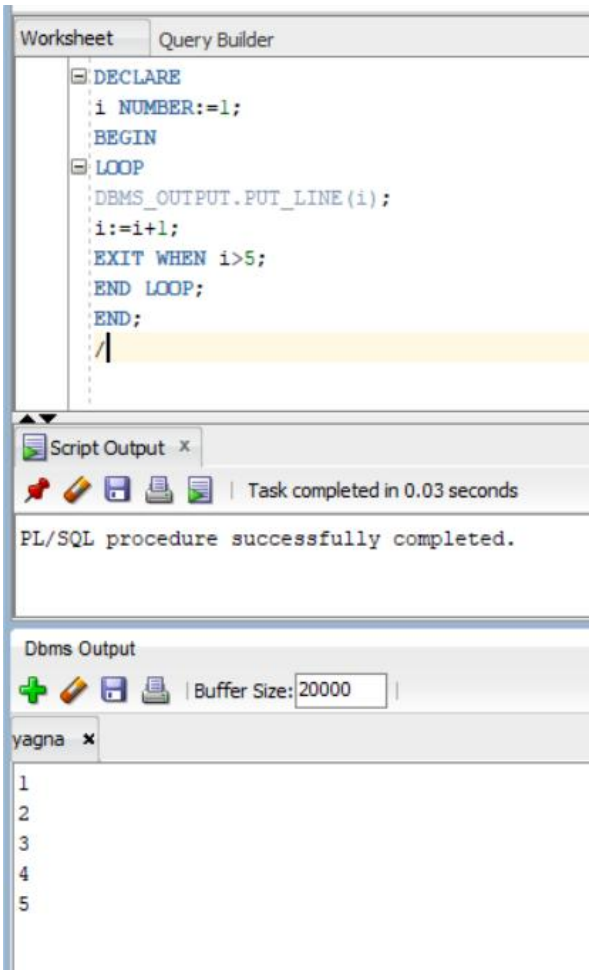
9. For loop



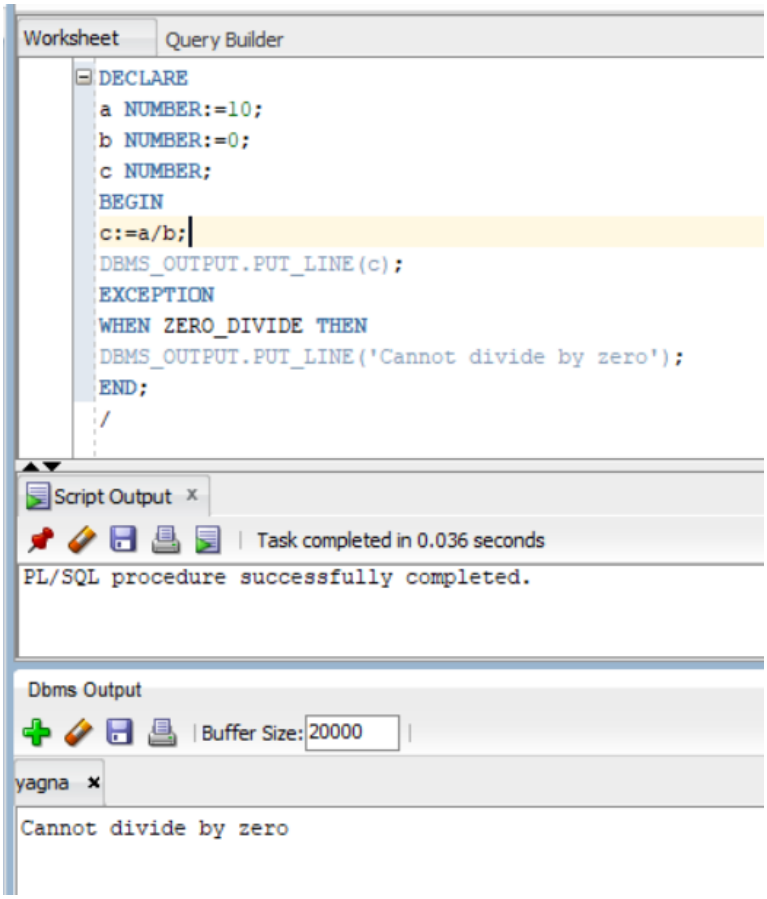
10. While loop



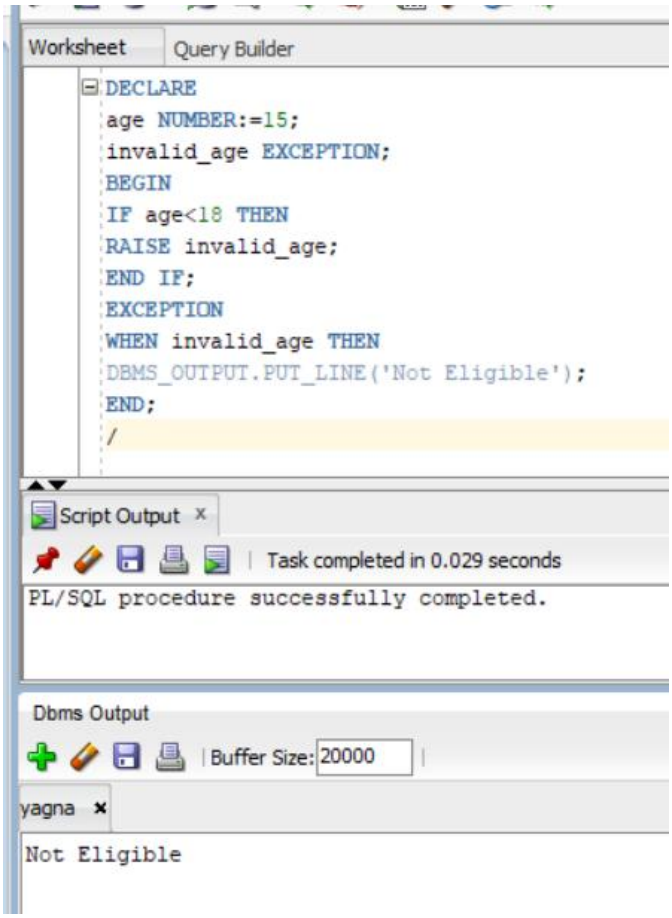
11. Simple loop



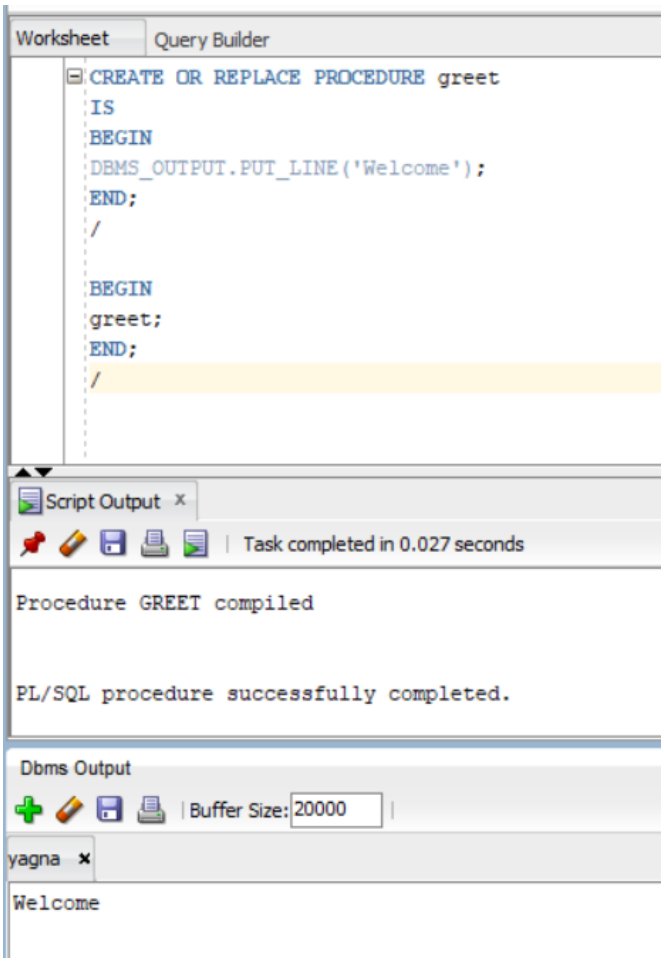
12. Predefined exception



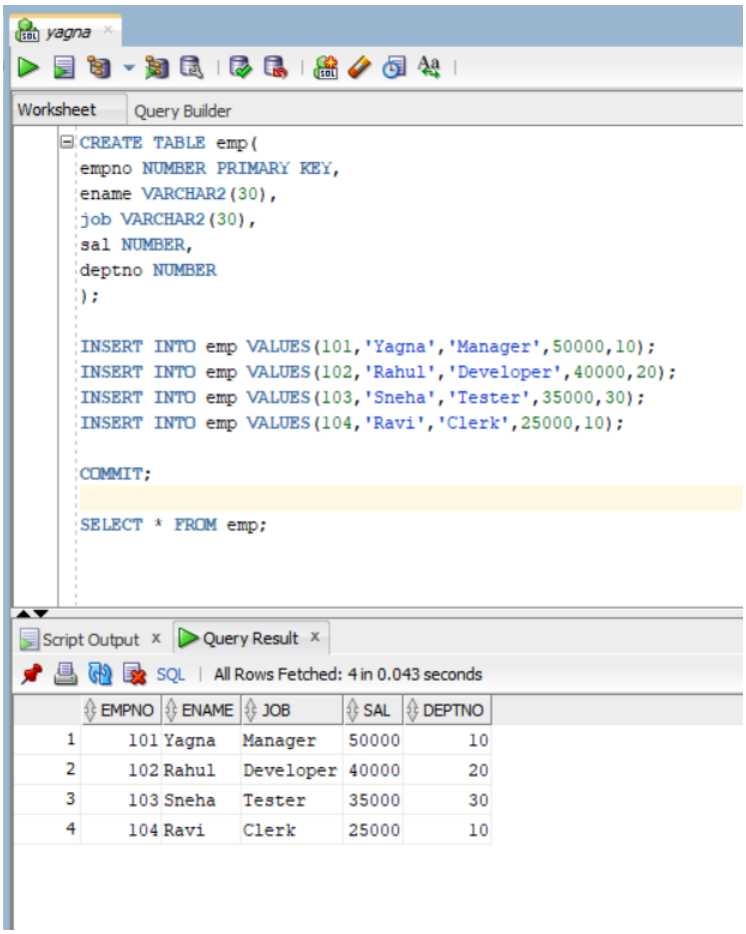
13. User defined exception



14. Stored procedure



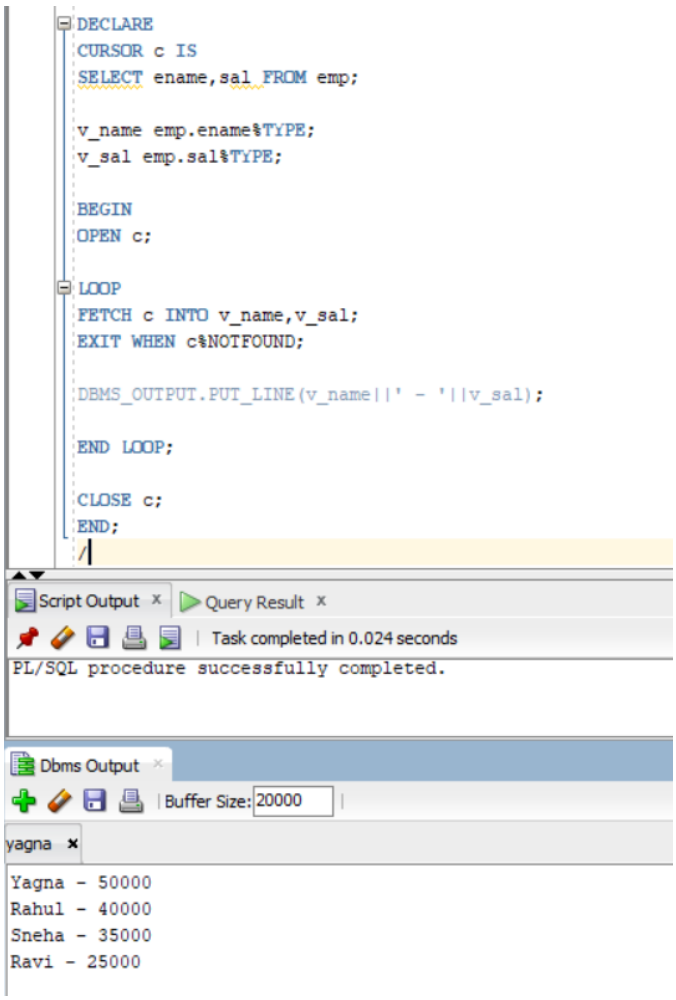
15. Create table



```
CREATE TABLE emp(
empno NUMBER PRIMARY KEY,
ename VARCHAR2(30),
job VARCHAR2(30),
sal NUMBER,
deptno NUMBER
);

INSERT INTO emp VALUES(101,'Yagna','Manager',50000,10);
INSERT INTO emp VALUES(102,'Rahul','Developer',40000,20);
INSERT INTO emp VALUES(103,'Sneha','Tester',35000,30);
INSERT INTO emp VALUES(104,'Ravi','Clerk',25000,10);

COMMIT;

SELECT * FROM emp;
```

Script Output | Query Result — All Rows Fetched: 4 in 0.043 seconds

	EMPNO	ENAME	JOB	SAL	DEPTNO
1	101	Yagna	Manager	50000	10
2	102	Rahul	Developer	40000	20
3	103	Sneha	Tester	35000	30
4	104	Ravi	Clerk	25000	10



```
DECLARE
CURSOR c IS
SELECT ename,sal FROM emp;

v_name emp.ename%TYPE;
v_sal emp.sal%TYPE;

BEGIN
OPEN c;

LOOP
FETCH c INTO v_name,v_sal;
EXIT WHEN c%NOTFOUND;

DBMS_OUTPUT.PUT_LINE(v_name||' - '||v_sal);

END LOOP;

CLOSE c;
END;
/
```

Script Output | Query Result — Task completed in 0.024 seconds

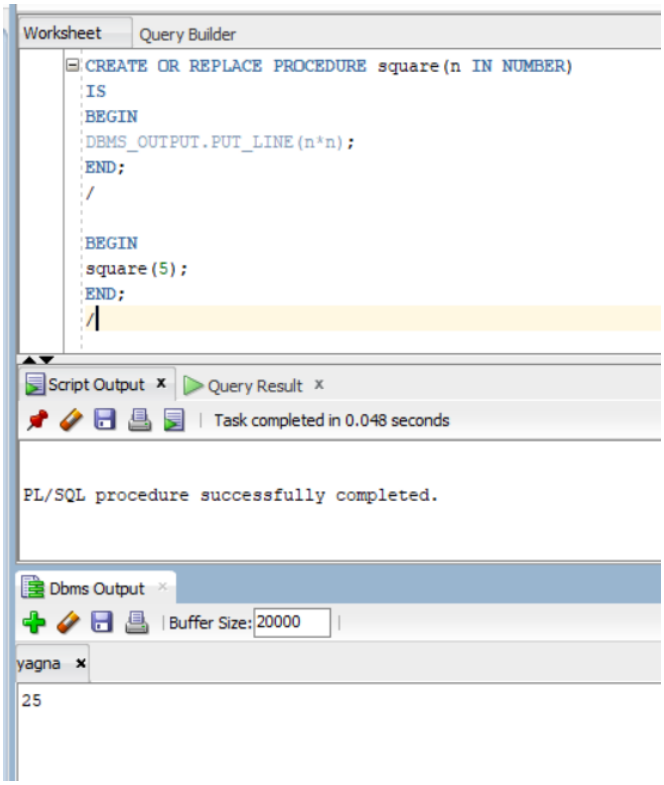
```
PL/SQL procedure successfully completed.
```

Dbms Output — Buffer Size: 20000

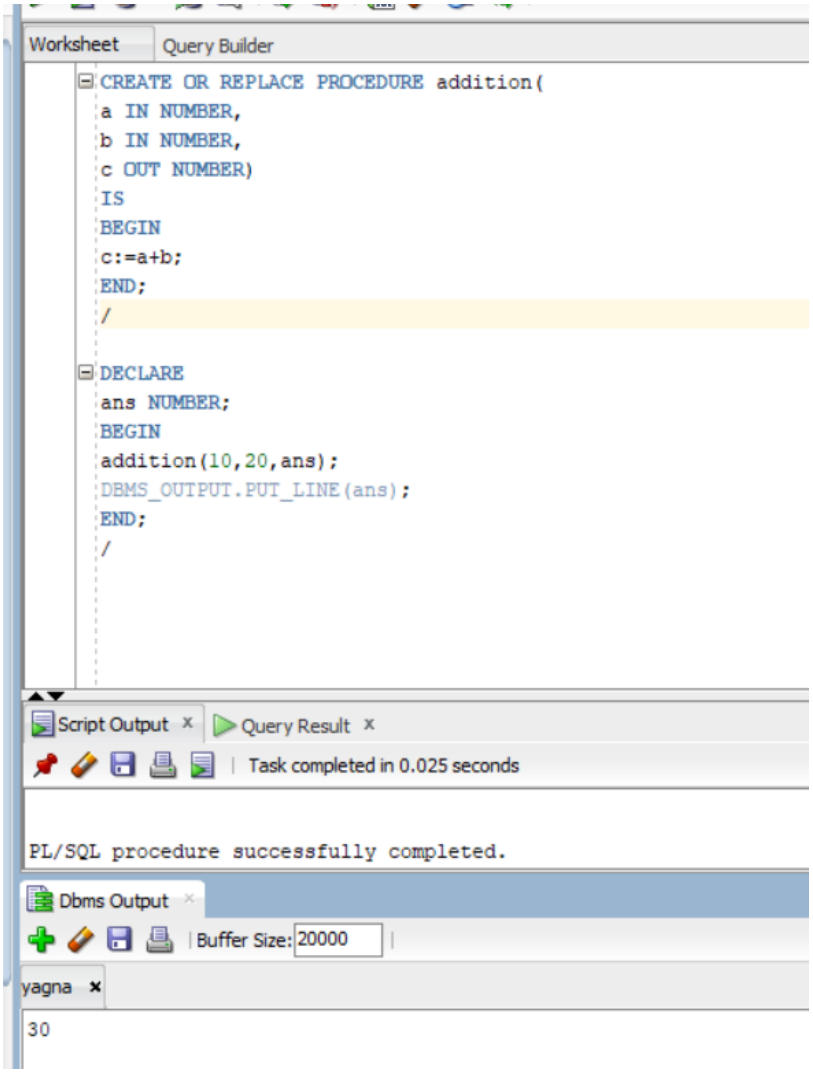
yagna

```
Yagna - 50000
Rahul - 40000
Sneha - 35000
Ravi - 25000
```

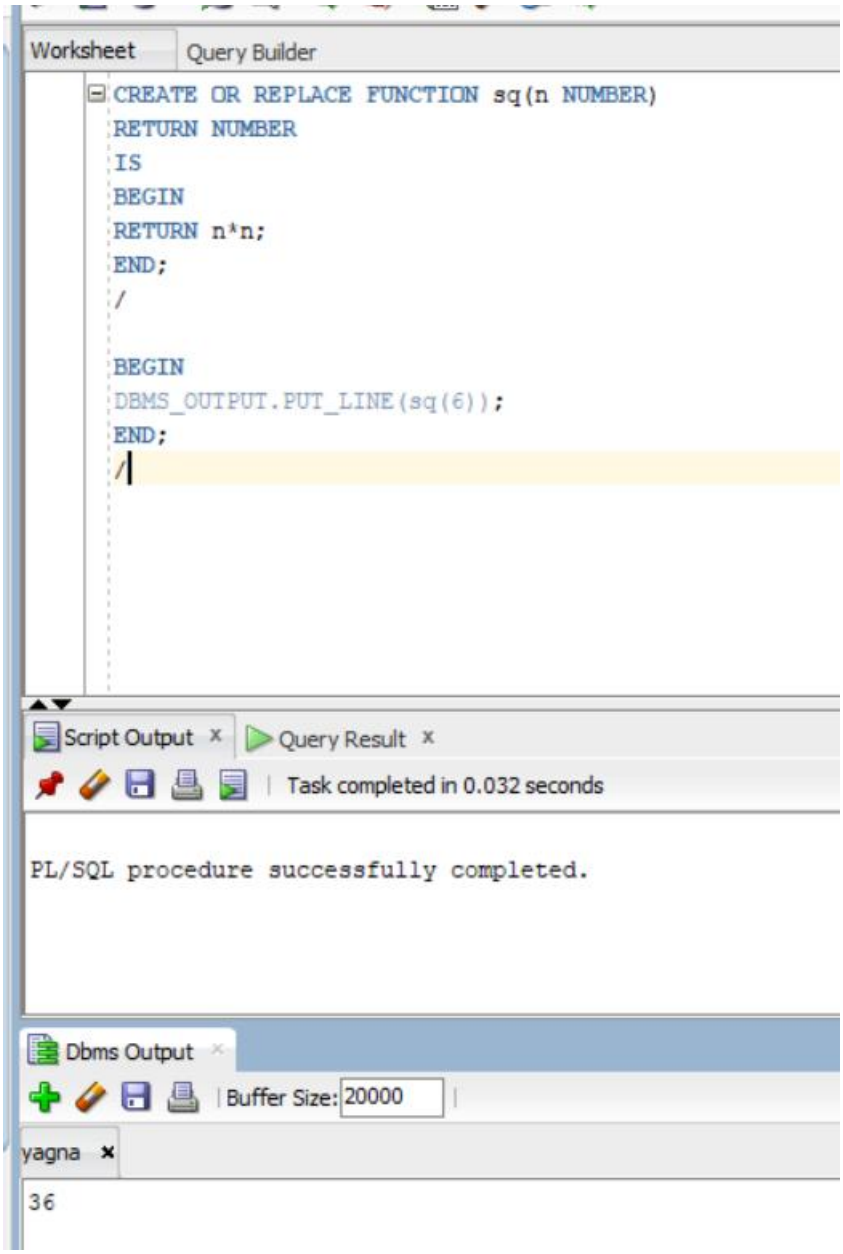
16. Procedure with in parameter



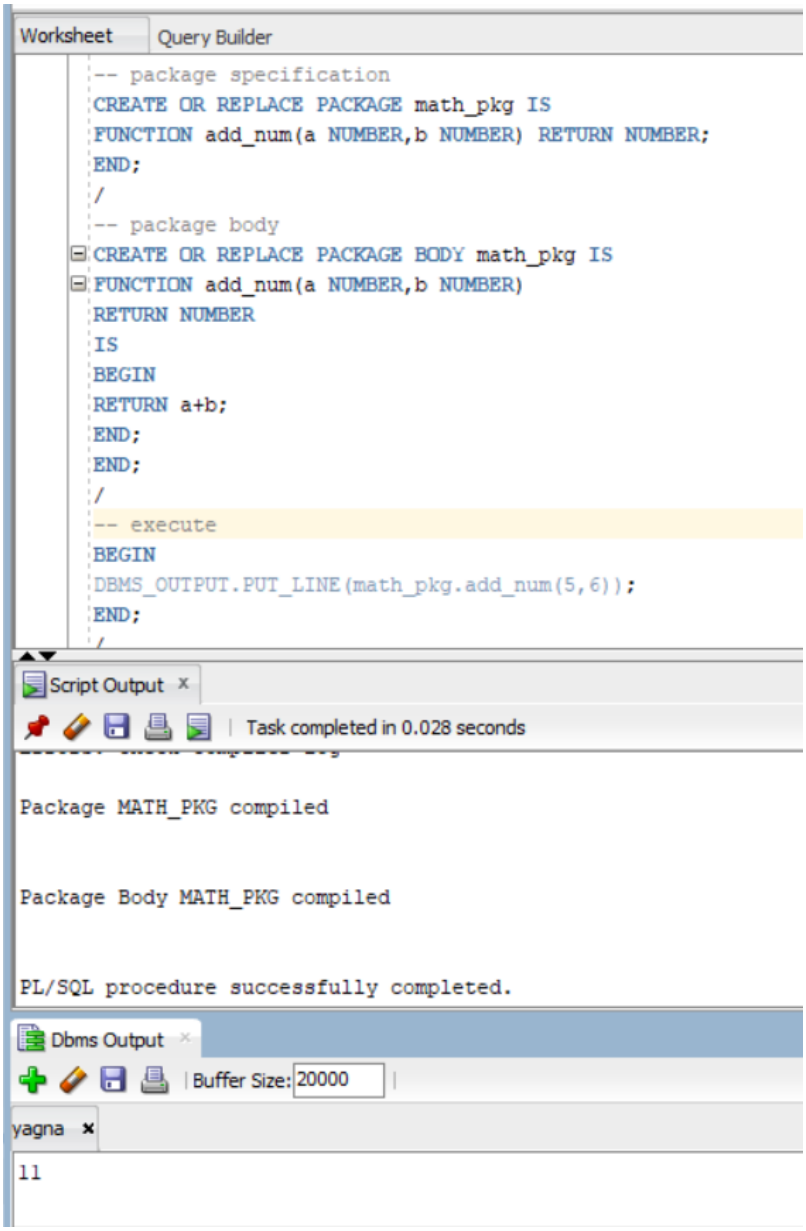
17. Procedure with out parameter



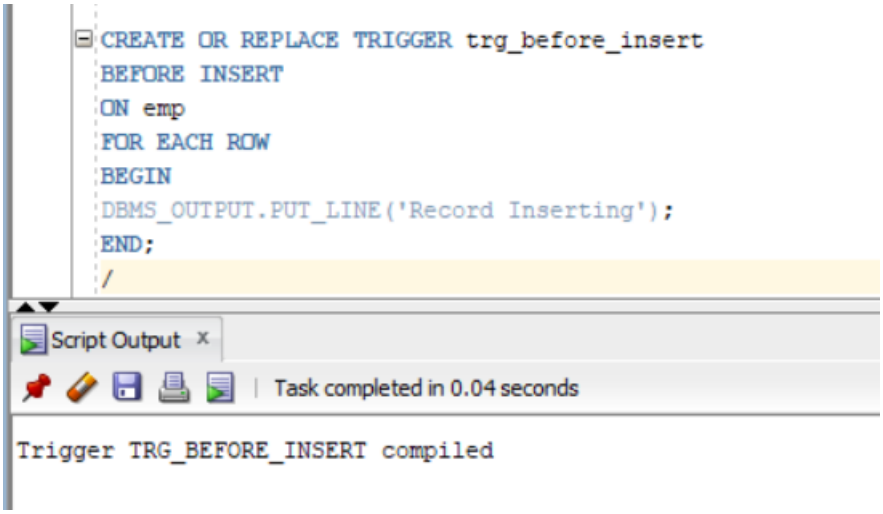
18. Function


Worksheet
Query Builder
CREATE OR REPLACE FUNCTION sq(n NUMBER)
RETURN NUMBER
IS
BEGIN
RETURN n*n;
END;
/
BEGIN
DBMS_OUTPUT.PUT_LINE(sq(6));
END;
/
Script Output
Query Result
Task completed in 0.032 seconds
PL/SQL procedure successfully completed.
Dbms Output
Buffer Size: 20000
yagna
36

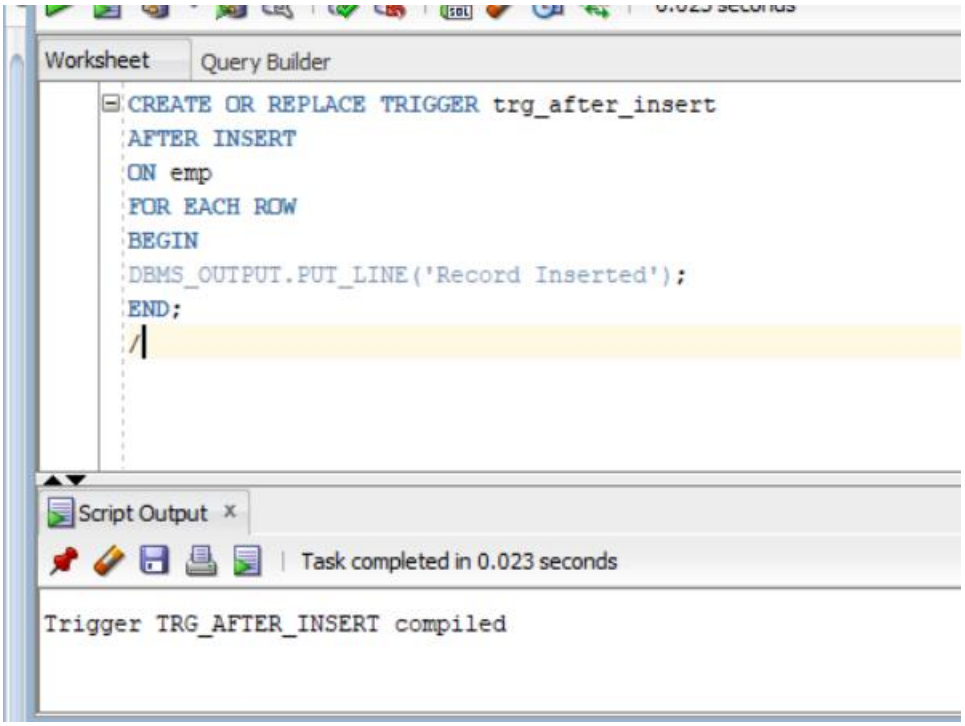
19. Package



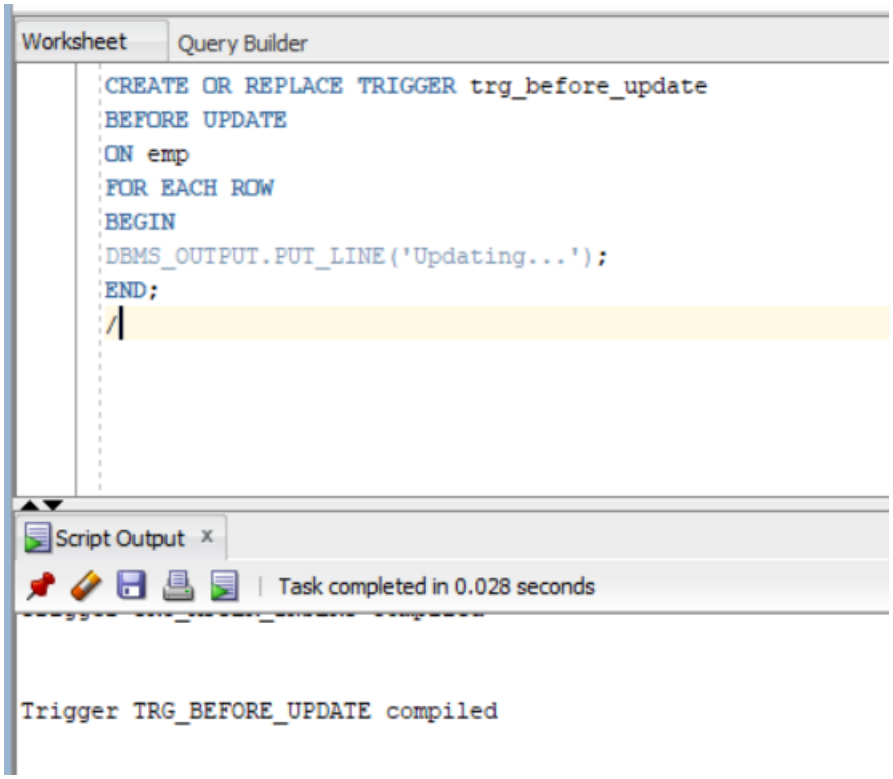
20. BEFORE INSERT trigger



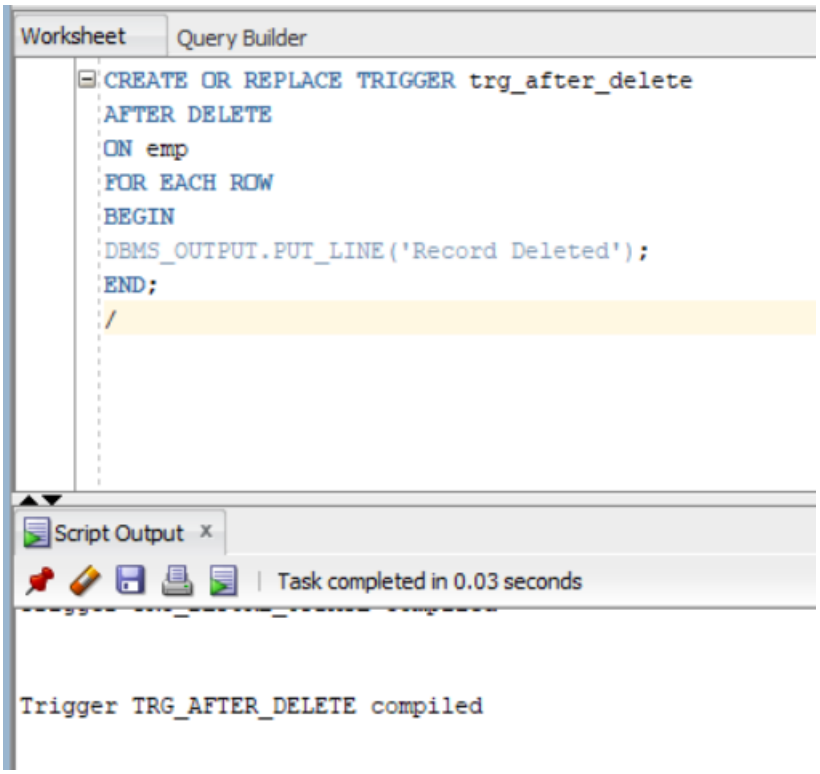
21. AFTER INSERT trigger



22. BEFORE UPDATE Trigger



23. AFTER DELETE Trigger



24. INSTEAD OF Trigger (on View)